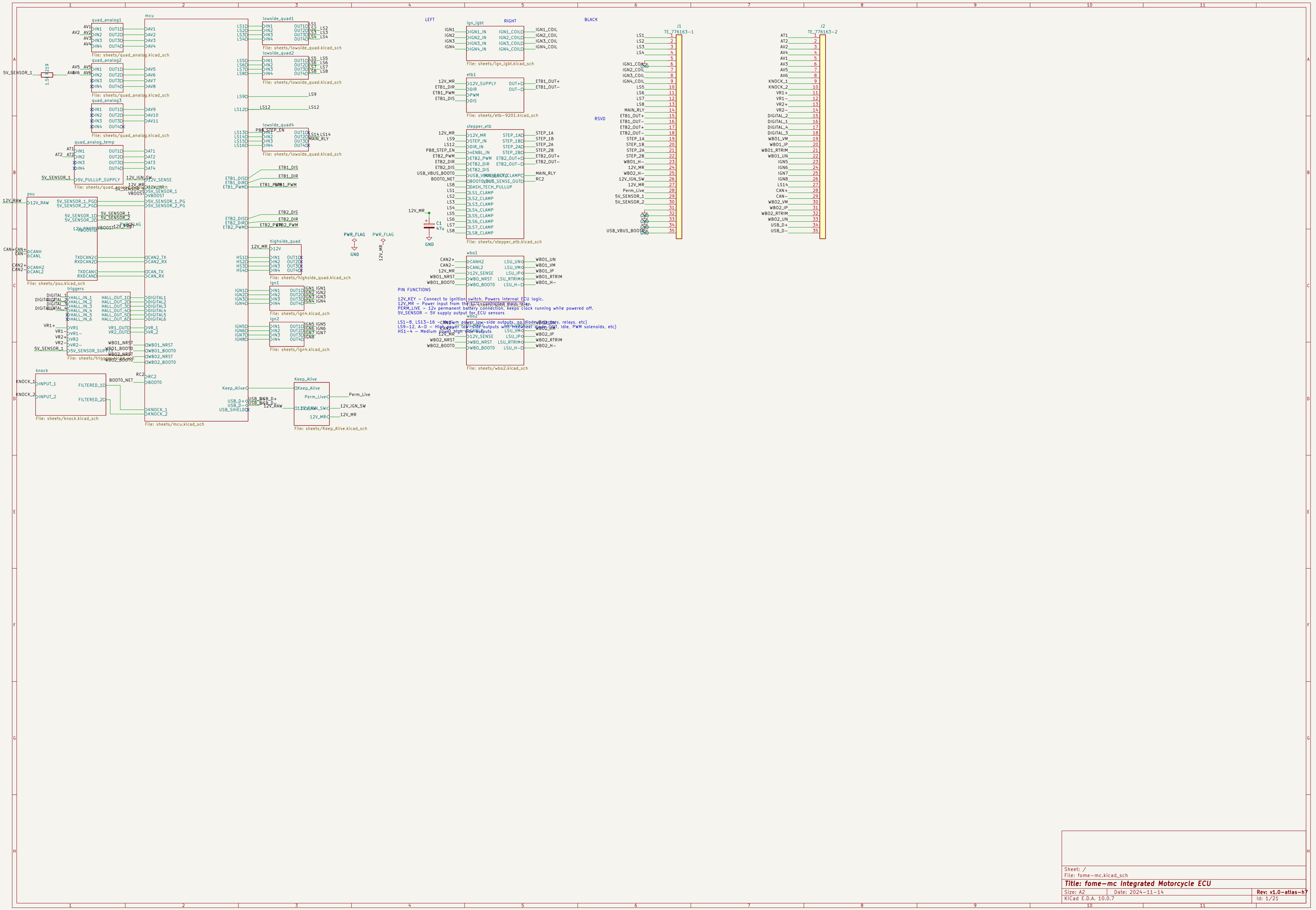
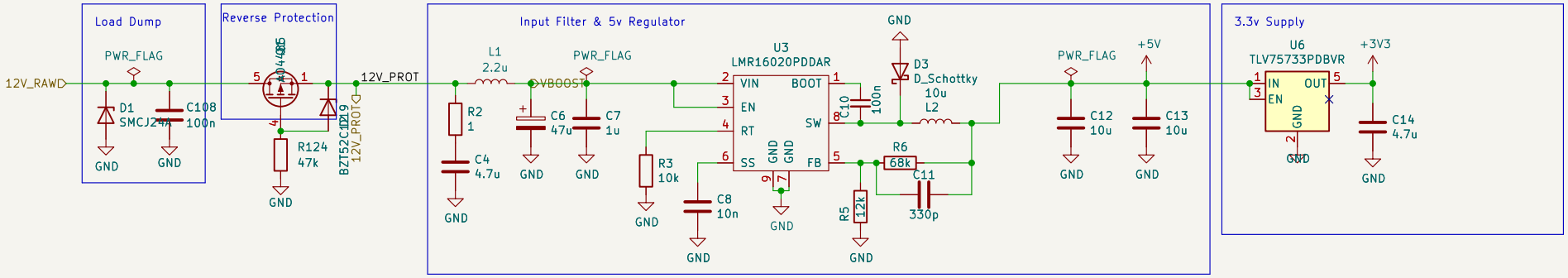
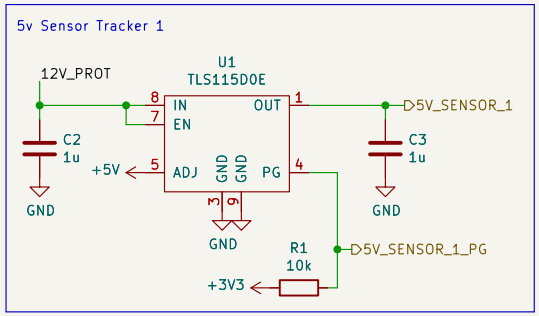


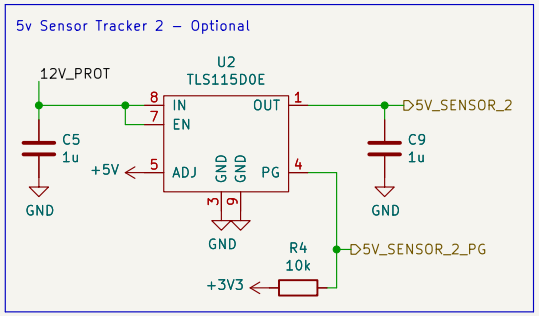
Sheet: /
File: fome-mc.kicad_sch

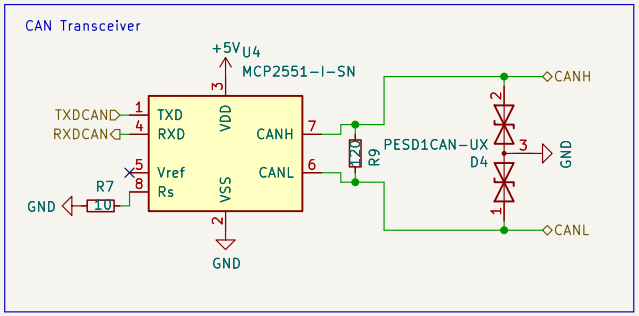
Title: fome-mc Integrated Motorcycle ECU

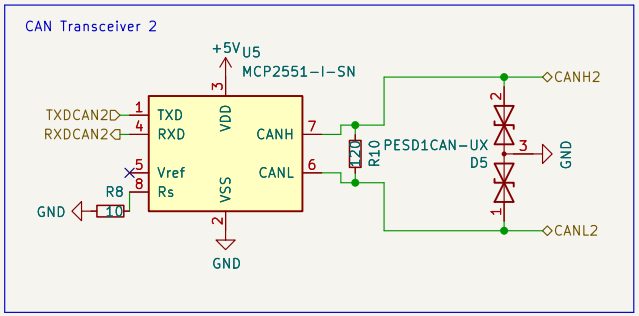
Size: A2	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 1/21









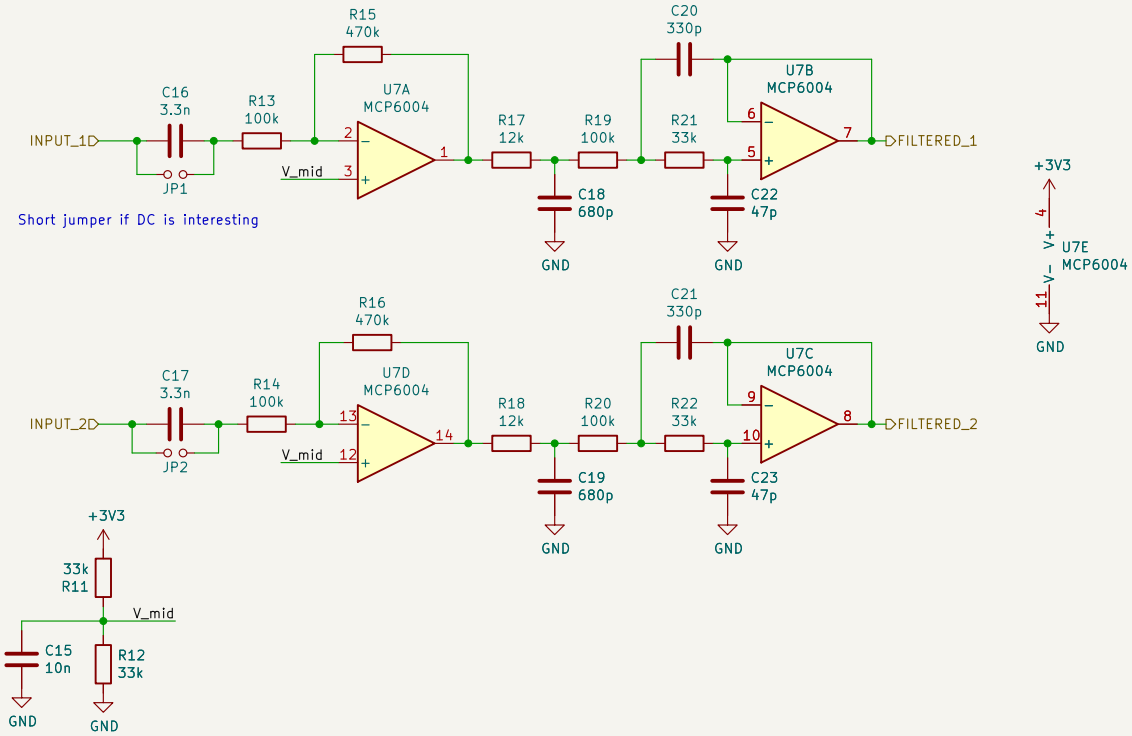


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Sheet: /psu/
File: psu.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 2/21

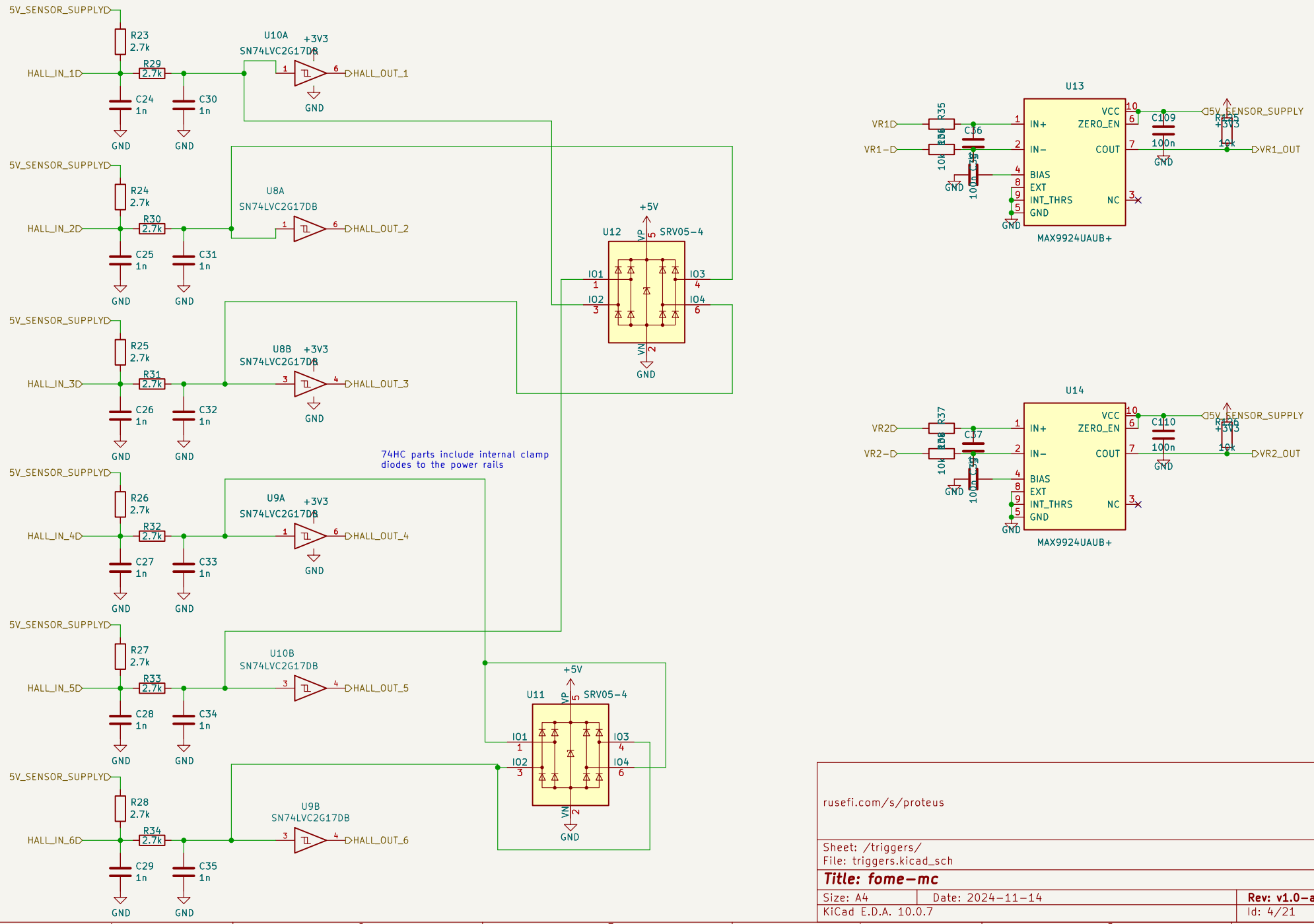


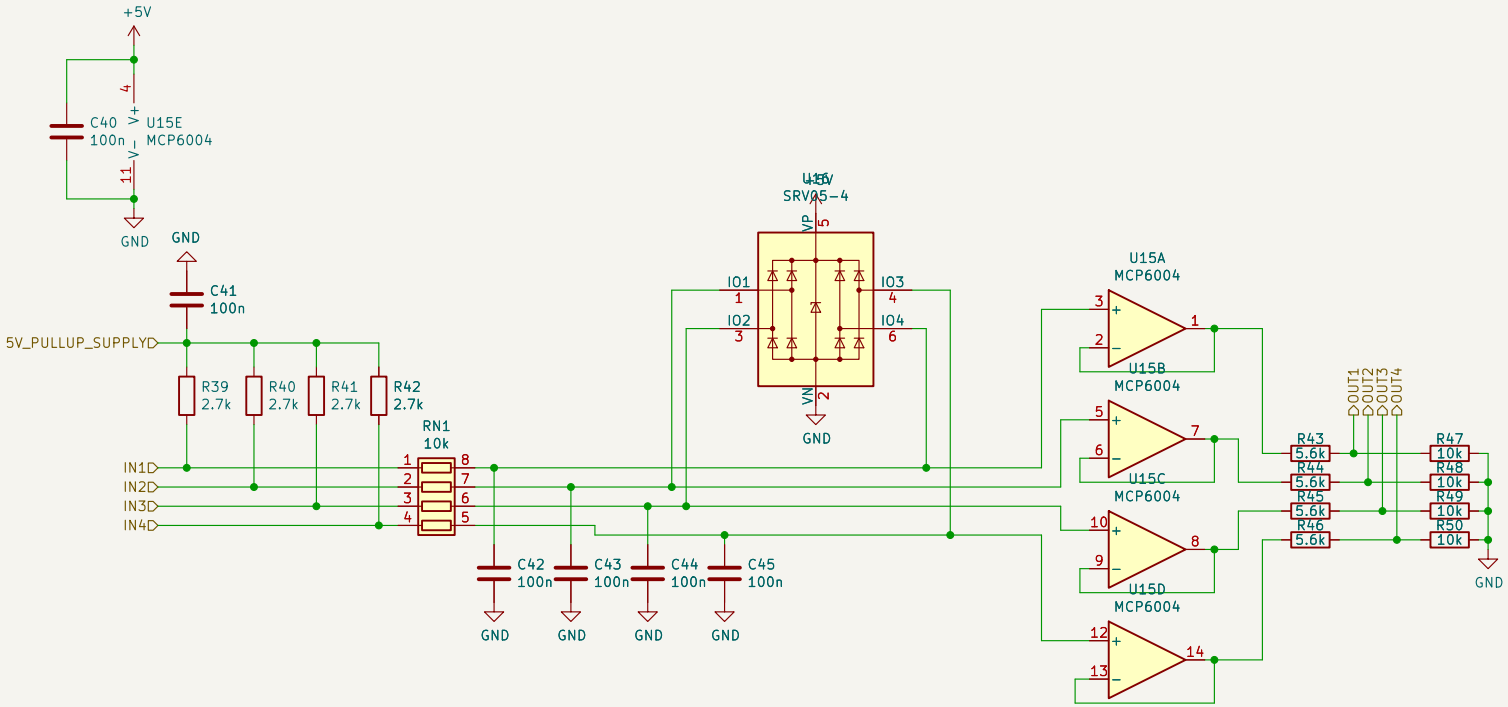
rusefi.com/s/proteus

Sheet: /knock/
File: knock.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 3/21


5V_SENSOR_SUPPLY
HALL_IN_1
HALL_IN_2
HALL_IN_3
HALL_IN_4
HALL_IN_5
HALL_IN_6
R23 2.7k
R29 2.7k
R24 2.7k
R30 2.7k
R25 2.7k
R31 2.7k
R26 2.7k
R32 2.7k
R27 2.7k
R33 2.7k
R28 2.7k
R34 2.7k
C24 1n
C30 1n
C25 1n
C31 1n
C26 1n
C32 1n
C27 1n
C33 1n
C28 1n
C34 1n
C29 1n
C35 1n
GND
U10A
SN74LVC2G17DB
+3V3
HALL_OUT_1
U8A
HALL_OUT_2
U8B
HALL_OUT_3
U9A
HALL_OUT_4
U10B
HALL_OUT_5
U9B
HALL_OUT_6
74HC parts include internal clamp diodes to the power rails
U12
SRV05-4
+5V
IO1
IO2
IO3
IO4
VP
VN
U11
U13
MAX9924UAUB+
VR1
VR2
R35
R36
R37
R38
C36
C37
C38
C39
10k
100n
IN+
IN−
VCC
ZERO_EN
COUT
BIAS
EXT
INT_THRS
NC
C109
C110
R39
R40
10k
VR1_OUT
VR2_OUT
rusefi.com/s/proteus
Sheet: /triggers/
File: triggers.kicad_sch
Title: fome-mc
Size: A4
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 4/21

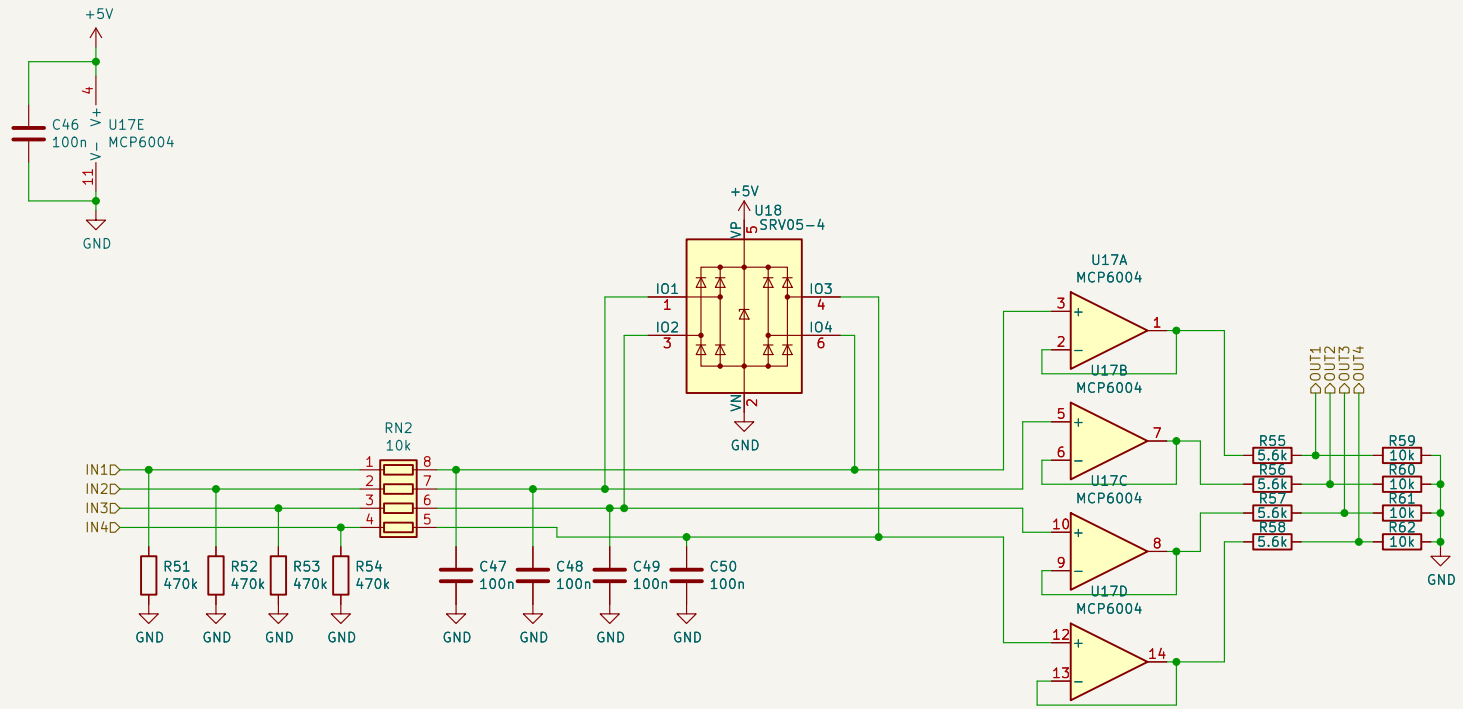


rusefi.com/s/proteus

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 5/21

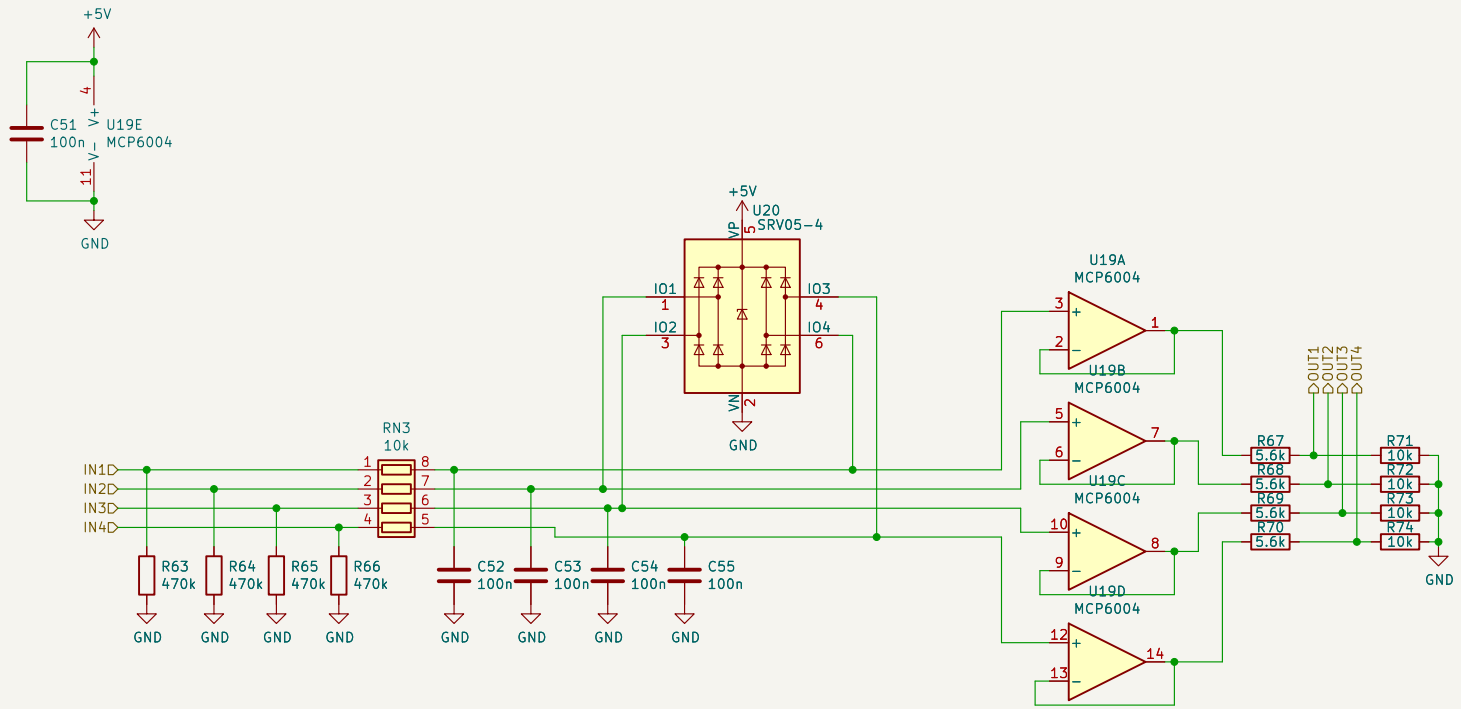


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Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 6/21

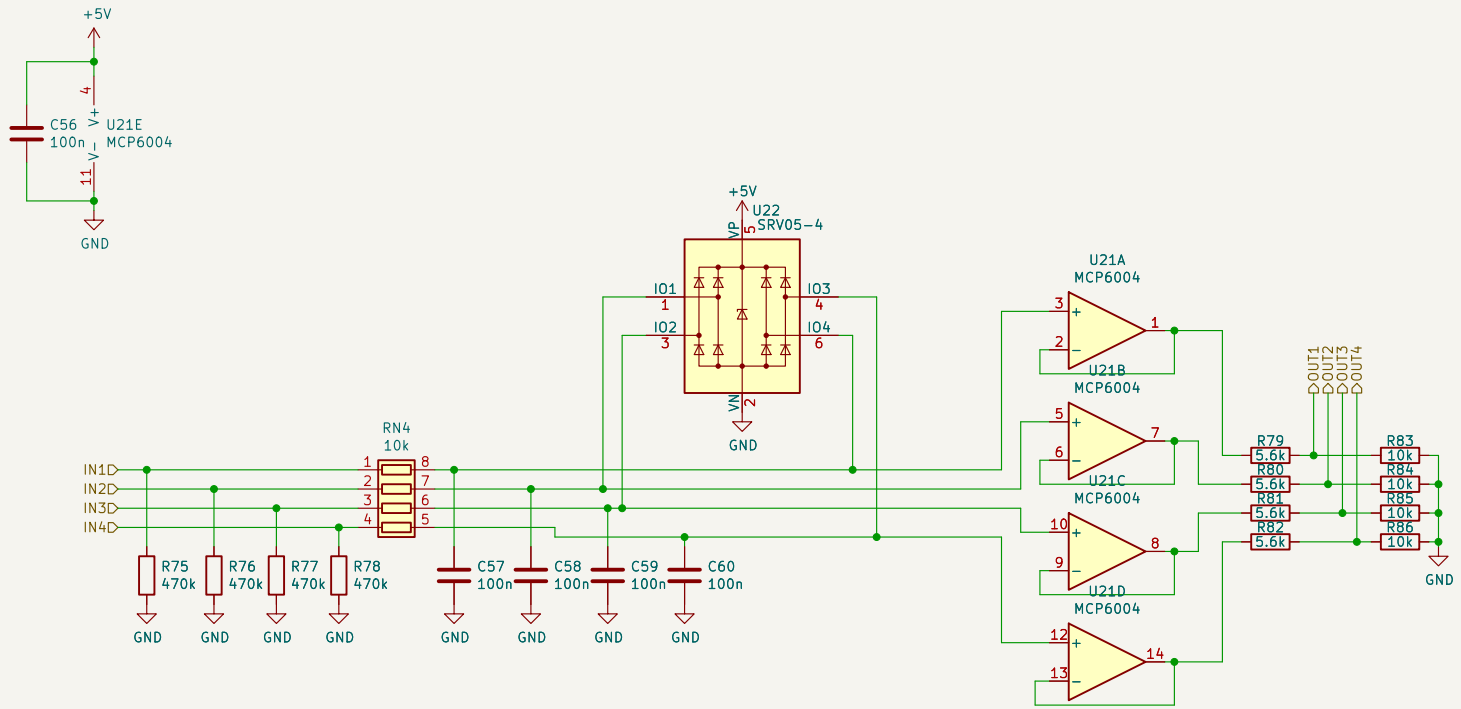


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Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 7/21

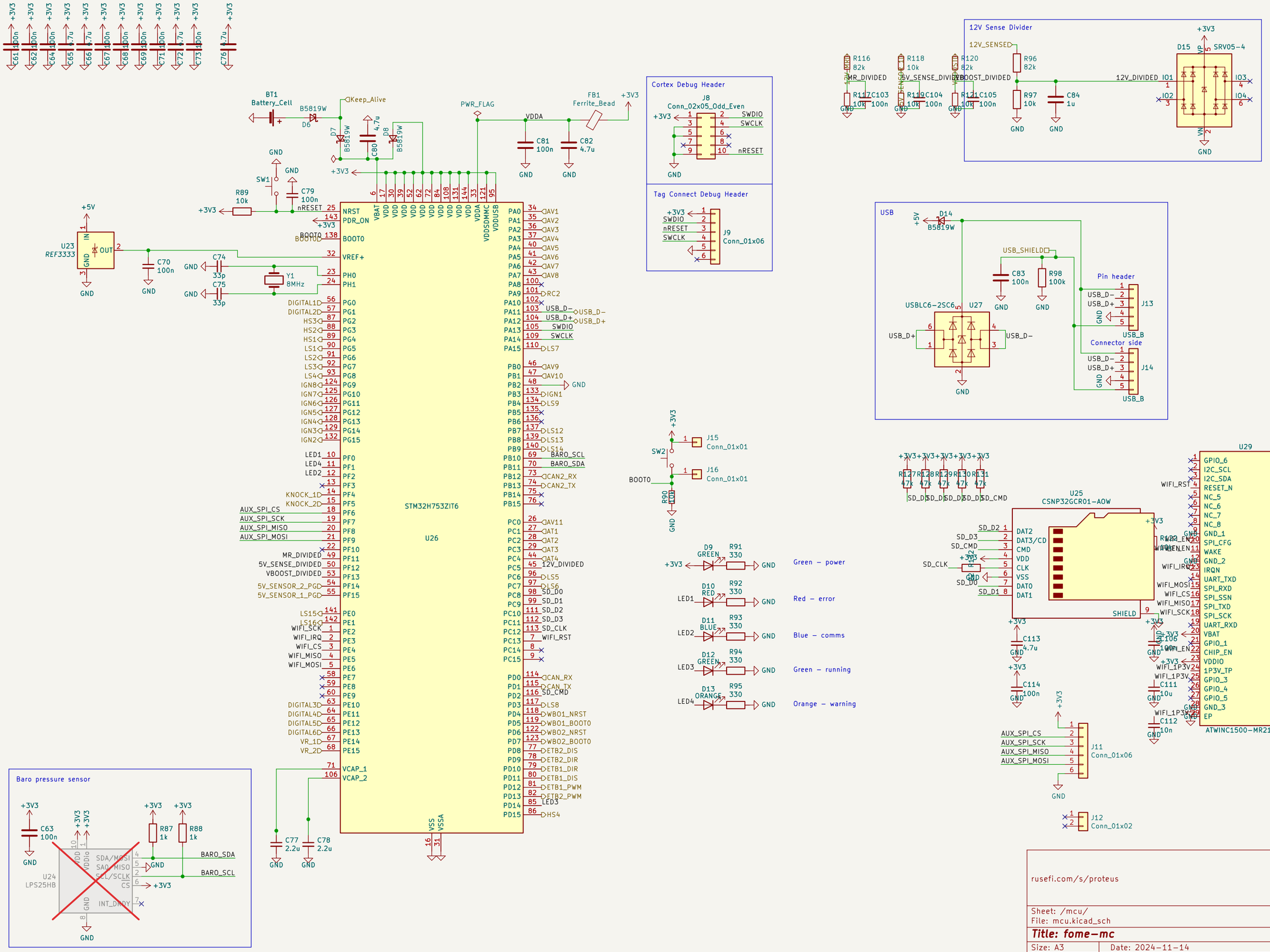


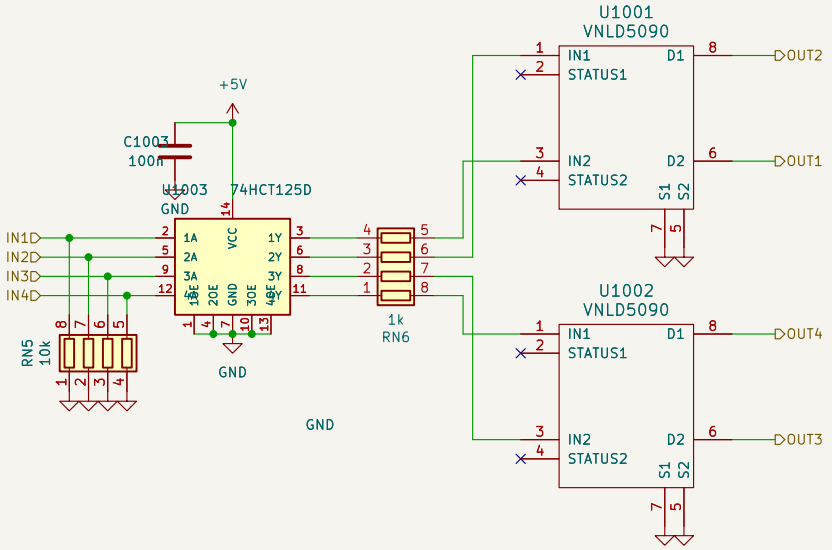
rusefi.com/s/proteus

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 8/21


C61 100n
C62 100n
C63 100n
C64 100n
C65 4.7u
C66 4.7u
C67 100n
C68 100n
C69 100n
C71 100n
C72 4.7u
C73 100n
C76 4.7u
+3V3
BT1
Battery_Cell
B5819W
D6
Keep_Alive
D7
D8
C80
4.7u
PWR_FLAG
VDDA
FB1
Ferrite_Bead
C81
100n
C82
4.7u
GND
SW1
C79
100n
nRESET
R89
10k
+5V
U23
REF3333
IN
OUT
C70
100n
C74
33p
C75
33p
Y1
8MHz
BOOT0
PDR_ON
VREF+
PH0
PH1
STM32H753ZIT6
U26
NRST
BOOT0
VBAT
VDD
VDDA
VDDSDMMC
VDDUSB
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PF0
PF1
PF2
PF3
PF4
PF5
PF6
PF7
PF8
PF9
PF10
PF11
PF12
PF13
PF14
PF15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PG0
PG1
PG2
PG3
PG4
PG5
PG6
PG7
PG8
PG9
PG10
PG11
PG12
PG13
PG14
PG15
VCAP_1
VCAP_2
C77
2.2u
C78
2.2u
Cortex Debug Header
J8
Conn_02x05_Odd_Even
SWDIO
SWCLK
nRESET
Tag Connect Debug Header
J9
Conn_01x06
12V Sense Divider
12V_SENSED
R116
82k
MR_DIVIDED
R117
10k
C103
100n
R118
10k
5V_SENSE_DIVIDED
R119
10k
C104
100n
R120
82k
VBOOST_DIVIDED
R121
10k
C105
100n
R96
82k
R97
10k
C84
1u
12V_DIVIDED
D15
SRV05-4
IO1
IO2
IO3
IO4
USB
D14
B5819W
USB_SHIELD
C83
100n
R98
100k
USBLC6-2SC6
U27
USB_D+
USB_D-
Pin header
J13
USB_B
Connector side
J14
J15
Conn_01x01
SW2
J16
BOOT0
R90
100k
D9
GREEN
R91
330
Green – power
D10
RED
R92
330
LED1
Red – error
D11
BLUE
R93
330
LED2
Blue – comms
D12
GREEN
R94
330
LED3
Green – running
D13
ORANGE
R95
330
LED4
Orange – warning
R127
47k
R128
47k
R129
47k
R130
47k
R131
47k
SD_D0
SD_D1
SD_D2
SD_D3
SD_CMD
SD_CLK
U25
CSNP32GCR01-AOW
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
C113
4.7u
C114
100n
U29
ATWINC1500-MR210UB
GPIO_6
I2C_SCL
I2C_SDA
RESET_N
NC_5
NC_6
NC_7
NC_8
GND_1
SPI_CFG
WAKE
GND_2
IRQN
UART_TXD
SPI_RXD
SPI_SSN
SPI_TXD
SPI_SCK
UART_RXD
VBAT
GPIO_1
CHIP_EN
VDDIO
1P3V_TP
GPIO_3
GPIO_4
GPIO_5
GND_3
EP
C106
100n
C111
10u
C112
10n
J11
Conn_01x06
AUX_SPI_CS
AUX_SPI_SCK
AUX_SPI_MISO
AUX_SPI_MOSI
J12
Conn_01x02
Baro pressure sensor
U24
LPS25HB
SDA/MOSI
SAO/MISO
SCL/SCLK
CS
INT_DRDY
R87
1k
R88
1k
C63
100n
BARO_SDA
BARO_SCL
rusefi.com/s/proteus
Sheet: /mcu/
File: mcu.kicad_sch
Title: fome-mc
Size: A3
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 9/21



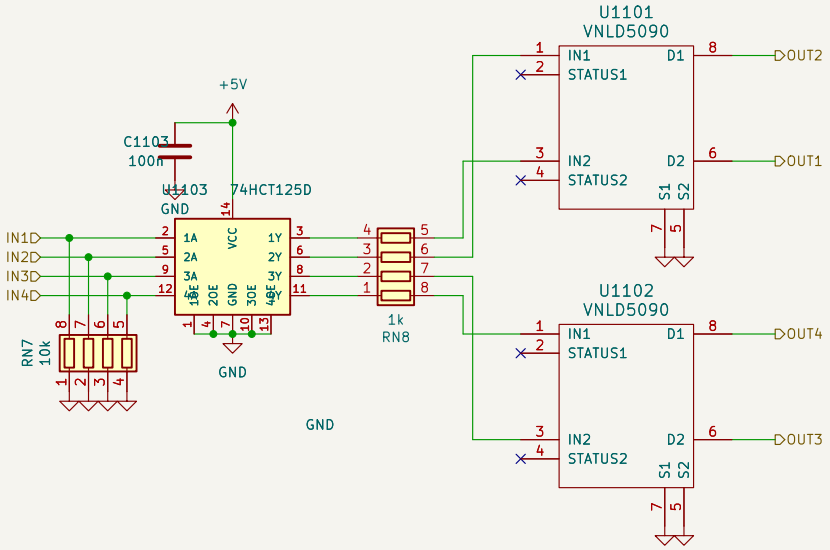
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 10/21



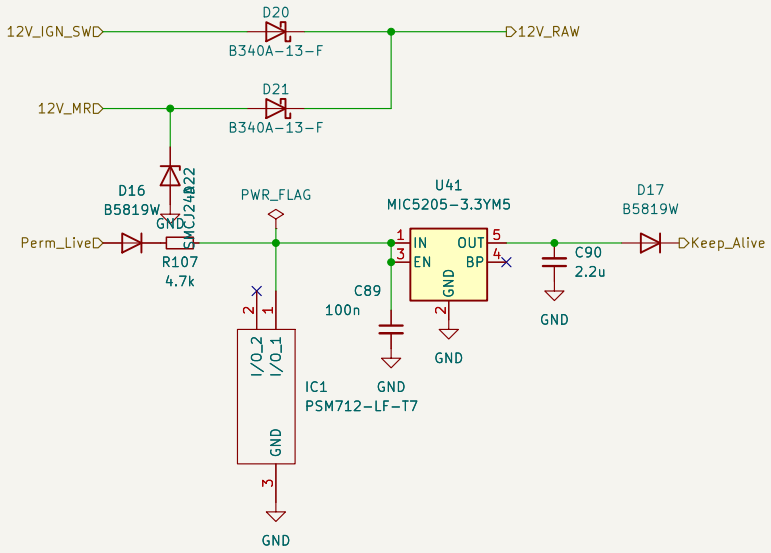
Locate output caps near connector pin

rusefi.com/s/proteus

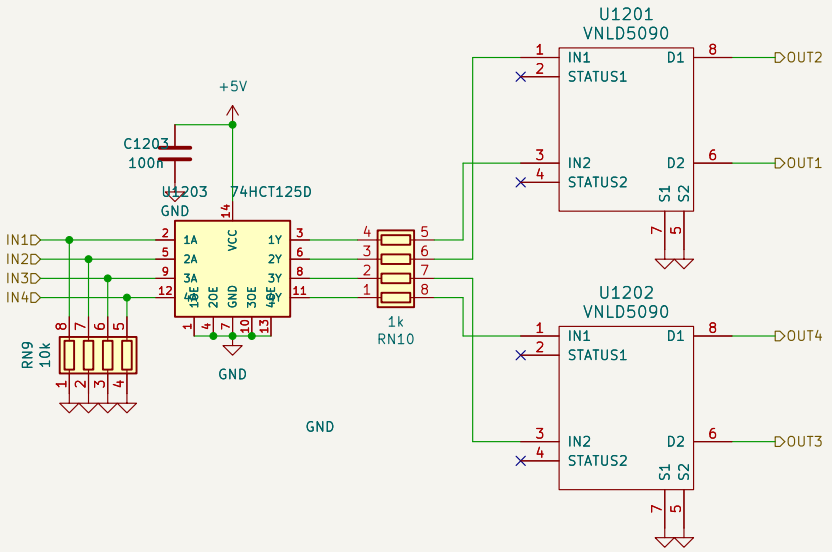
Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 11/21



Sheet: /Keep_Alive/		
File: Keep_Alive.kicad_sch		
Title: fome-mc		
Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 12/21



Locate output caps near connector pin

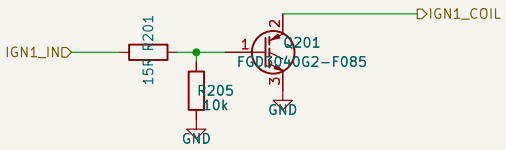
rusefi.com/s/proteus

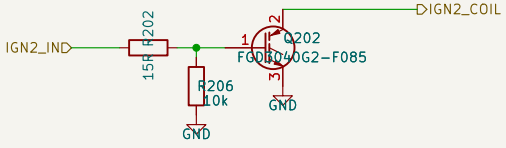
Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

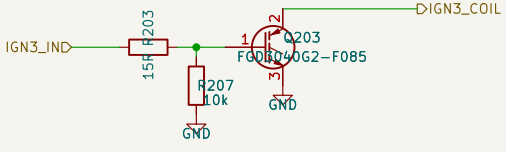
Title: fome-mc

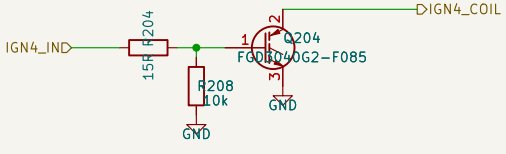
Size: A4 | Date: 2024-11-14 | **Rev: v1.0-atlas-h7**

KiCad E.D.A. 10.0.7 | Id: 13/21





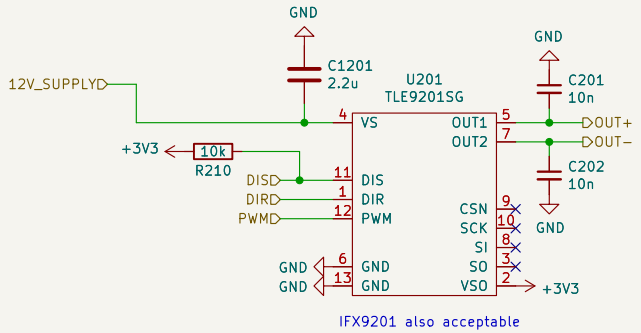




Sheet: /ign_igbt/
File: ign_igbt.kicad_sch

Title: fome-mc 4x High-Current IGBT Ignition Stage

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 14/21

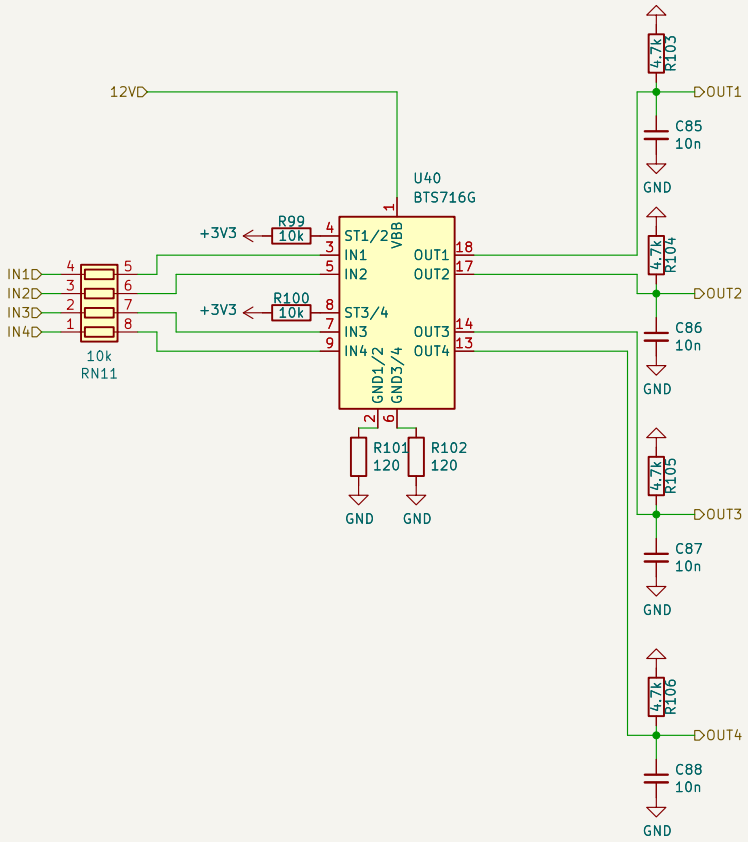


rusefi.com/s/proteus
github.com/mck1117/proteus
rusEFI

Sheet: /etb1/
File: etb-9201.kicad_sch

Title: rusEfi Proteus

Size: A4	Date: 2022-04-09	**Rev: v0.7**
KiCad E.D.A. 10.0.7		Id: 15/21

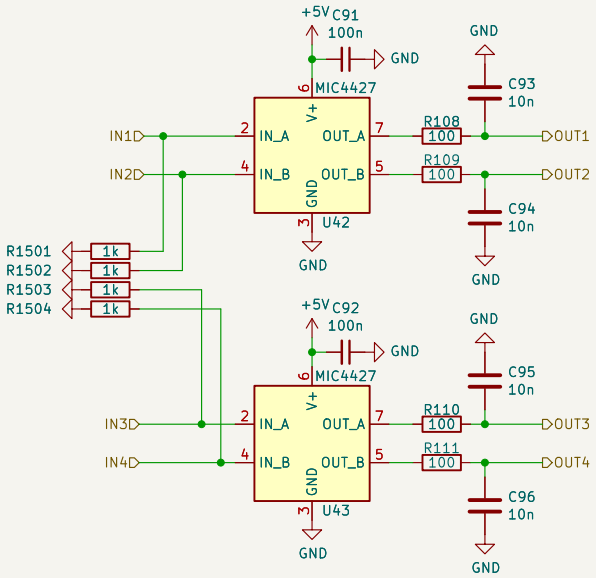


rusefi.com/s/proteus

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 16/21

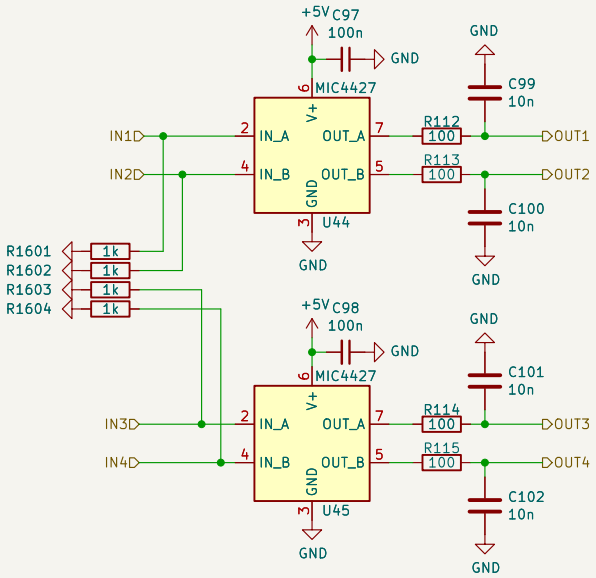


rusefi.com/s/proteus

Sheet: /ign1/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 17/21

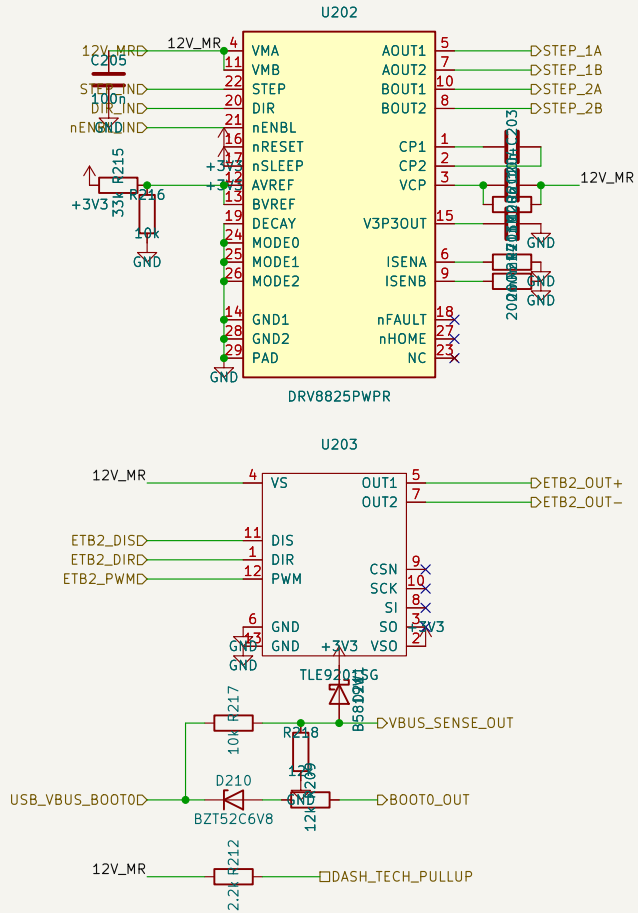


rusefi.com/s/proteus

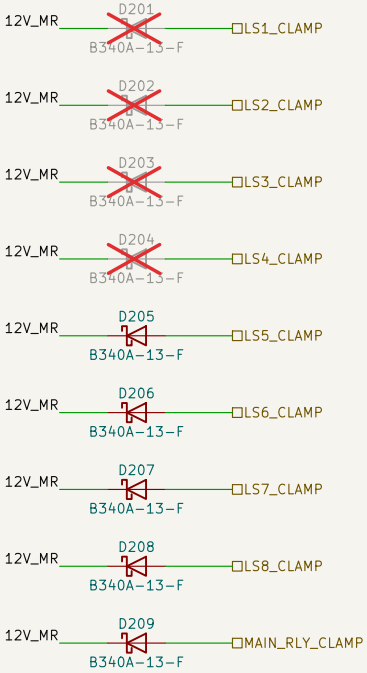
Sheet: /ign2/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 18/21



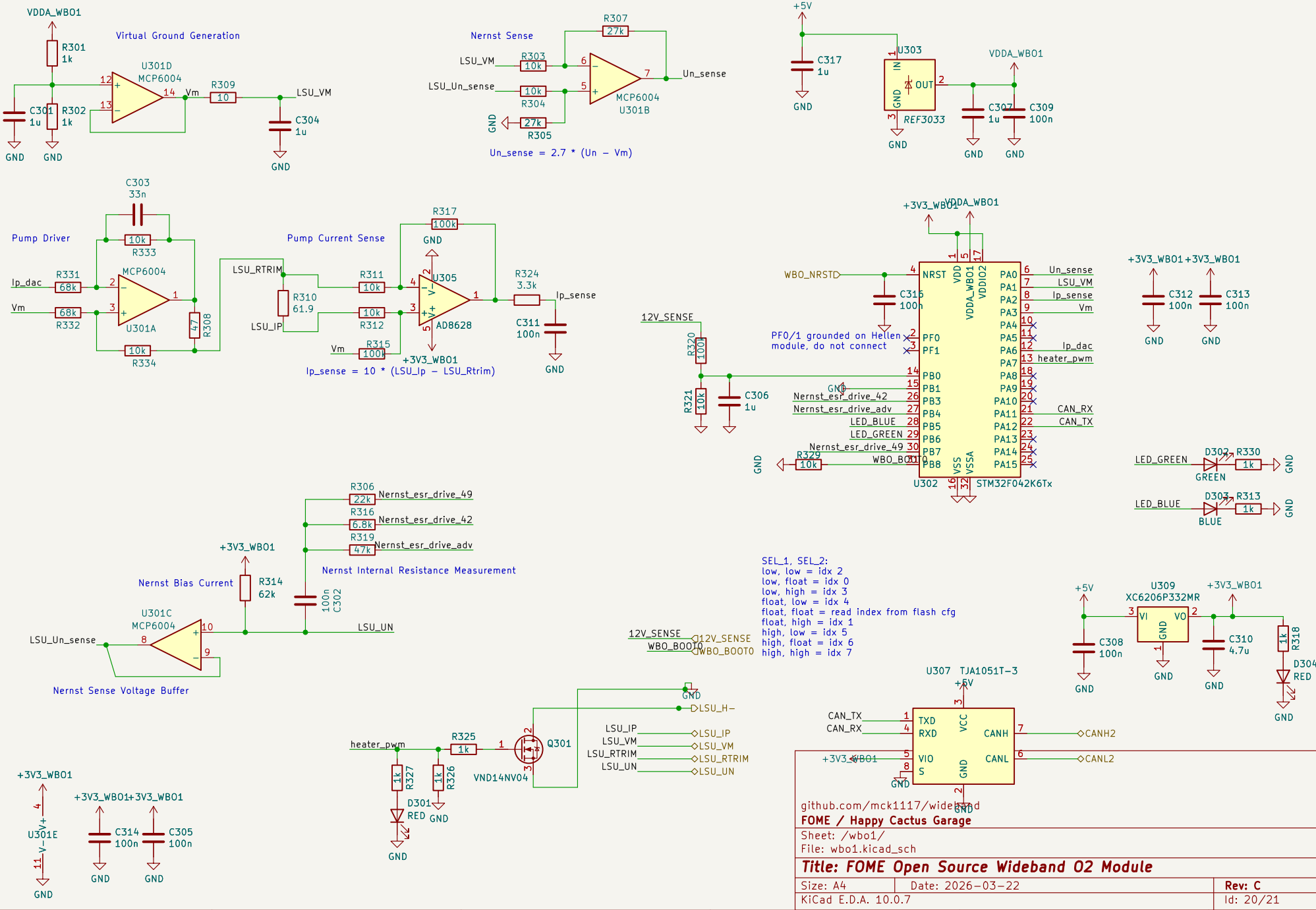
D201..D204: DNP on INJ1..4 (/LS1../LS4) so U1001/U1002 VNLD5090TR-E use internal 42V active clamp for fast injector turn-off.
D205..D209: Populated on LS5..LS8 & MAIN_RLY inductive relay/solenoid channels.

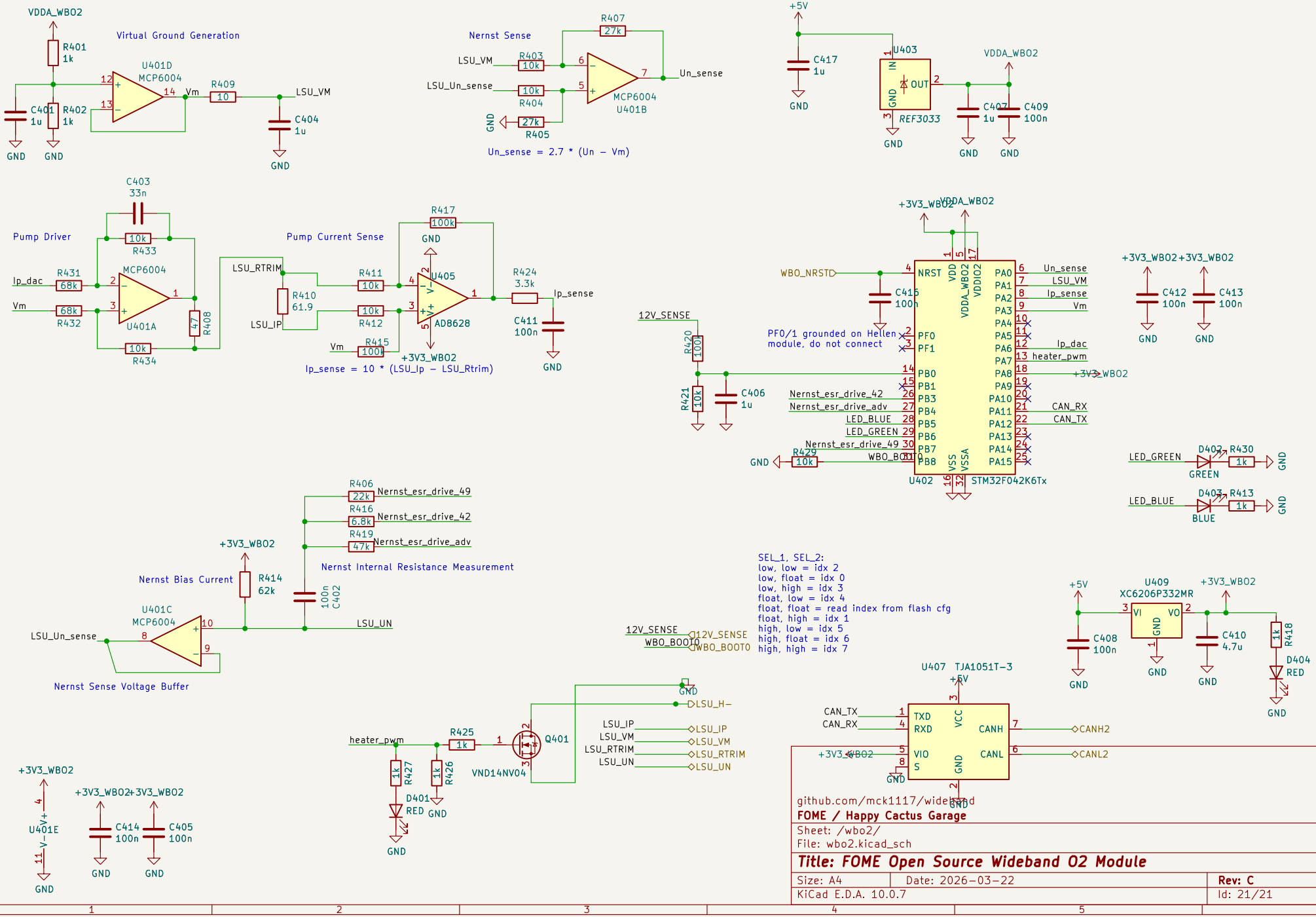


Sheet: /stepper_etb/
File: stepper_etb.kicad_sch

Title: fome-mc Stepper Driver + ETB2 H-Bridge + Low-Side Clamps + BOO

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 19/21


Virtual Ground Generation
Nernst Sense
Un_sense = 2.7 * (Un − Vm)
Pump Driver
Pump Current Sense
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
Nernst Internal Resistance Measurement
Nernst Bias Current
Nernst Sense Voltage Buffer
PF0/1 grounded on Hellen module, do not connect
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
github.com/mck1117/wideband
FOME / Happy Cactus Garage
Sheet: /wbo1/
File: wbo1.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 20/21


Virtual Ground Generation
Nernst Sense
Un_sense = 2.7 * (Un − Vm)
Pump Driver
Pump Current Sense
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
PF0/1 grounded on Hellen module, do not connect
Nernst Internal Resistance Measurement
Nernst Bias Current
Nernst Sense Voltage Buffer
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
github.com/mck1117/wideband
FOME / Happy Cactus Garage
Sheet: /wbo2/
File: wbo2.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 21/21